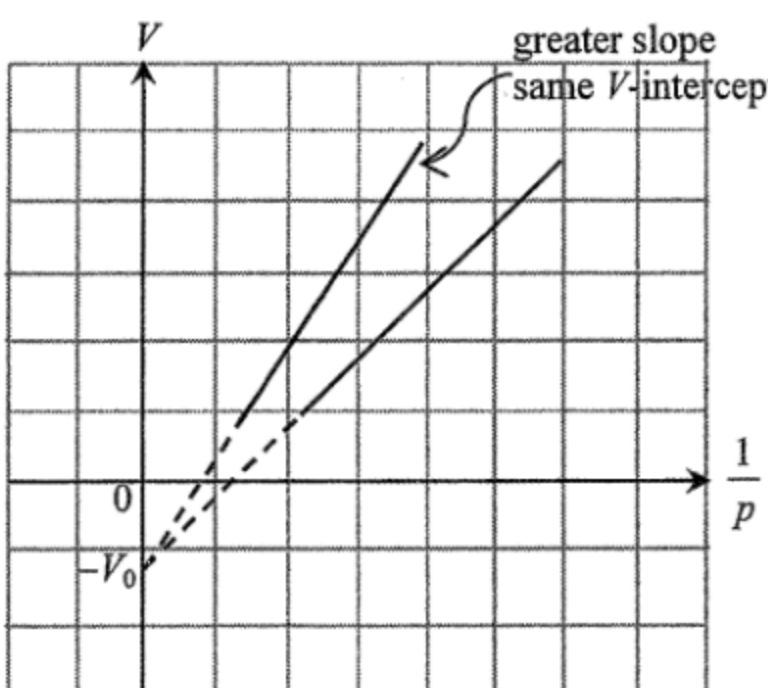


|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                    |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--|
| <p>1. (a) - Put the sphere into the water bath for a few minutes<br/> - Put / transfer the sphere into the polystyrene cup (of water)<br/> - Measure the final / maximum temperature <math>T_f</math> of the water with a thermometer</p> $0.80 \times c_b \times (80 - T_f) = 0.50 \times 4200 \times (T_f - T_0)$ $c_b = 2625 \times \frac{T_f - T_0}{80 - T_f} \text{ (J kg}^{-1} \text{ }^\circ\text{C}^{-1}\text{)}$ <p>Precautions:</p> <ul style="list-style-type: none"> <li>- Dry the sphere with the towel quickly before putting it into the cup</li> <li>- Make sure the sphere is fully immersed in water</li> <li>- Stir the water thoroughly</li> </ul> | <p>1A<br/>1A<br/>1A<br/><br/>1A<br/><br/><br/>1A<br/>Any<br/>TWO<br/>1A<br/>1A</p> |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 6                                                                                  |  |
| <p>(b) Thermal energy / heat is lost during the transfer / drying of the sphere</p> <div style="border: 1px solid black; padding: 5px;"> <p>Or Thermal energy / heat absorbed by thermometer, stirrer or cup</p> <p>Or The temperature of the sphere is higher than <math>T_f</math> when this final temperature is measured (i.e. <math>T_f</math> not reaching its maximum)</p> </div> <p>Thus temperature rise of water in the cup is lower than it should be.</p>                                                                                                                                                                                                  | <p>1A<br/><br/>1A<br/>1A<br/><br/>1A<br/><br/>2</p>                                |  |

|                                                                                                                                                                                                                                                                                                                                                                                            |                                           |                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|----------------------------|
| <p>2. (a) <math>pV = nRT</math><br/> <math>(1.0 \times 10^5)(6.0 \times 10^{-5}) = n(8.31)(273 + 25)</math><br/> <math>n = 2.422891 \times 10^{-3} \text{ moles} \approx 2.42 \times 10^{-3} \text{ moles}</math></p> <p>No. of molecules = <math>nN_A</math><br/> <math>= n \times 6.02 \times 10^{23}</math><br/> <math>= 1.458581 \times 10^{21} \approx 1.46 \times 10^{21}</math></p> | <p>1M<br/>1M<br/><br/>1A</p>              |                            |
| <p><i>Alternative method:</i></p> $pV = nRT = \left( \frac{N}{N_A} \right) RT \Rightarrow N = \left( \frac{pV}{RT} \right) N_A$ $N = \frac{(1.0 \times 10^5)(6.0 \times 10^{-5})}{(8.31)(273 + 25)} \times (6.02 \times 10^{23})$ $= 1.458581 \times 10^{21} \approx 1.46 \times 10^{21}$                                                                                                  | <p>1M<br/><br/>1M<br/><br/>1A</p>         |                            |
| <p>(b) (i) - The piston should be pushed or pulled slowly / lightly.<br/> - Avoid taking readings immediately after moving the piston.<br/> - The syringe should not be grasped by hand too long (when the piston is pushed in or pulled out).</p>                                                                                                                                         | <p>3<br/>1A<br/>1A<br/>1A<br/>Any ONE</p> |                            |
| <p>(ii) <math>V_0</math> - volume of gas trapped in the rubber tubing / space connecting the pressure sensor and syringe.</p>                                                                                                                                                                                                                                                              | <p>1<br/>1A</p>                           |                            |
| <p>(iii)</p>                                                                                                                                                                                                                                                                                           | <p>1<br/>1A<br/>1A<br/>2</p>              | <p>NO mark for a curve</p> |

3. (a)  $a = \frac{6-0}{2-0}$   
 $= 3 \text{ m s}^{-2} \text{ (downwards)}$

1M

1A

2

(b) A: 395 N      B: 569 N      C: 685 N

1A

In stage B, balance reading = weight (Newton's 1<sup>st</sup> law),

$$mg = m \times 9.81 \text{ m s}^{-2} = 569 \text{ N}$$

$$m = 58.0 \text{ kg}$$

1M

1A

For  $g = 10 \text{ m s}^{-2}$ ,  $m = 56.9 \text{ kg}$

Or In stage A, according to Newton's 2<sup>nd</sup> law  
 $(569 - 395) \text{ N} = ma = m(3 \text{ m s}^{-2})$   
 $m = 58.0 \text{ kg}$

1M

1A

3

(c) (i) For stage C, by Newton's 2<sup>nd</sup> law,

$$F = ma$$

$$(569 - 685) \text{ N} = (58.0 \text{ kg}) a$$

$$a = -2 \text{ m s}^{-2}$$

1M

For  $g = 10 \text{ m s}^{-2}$ ,  $m = 56.9 \text{ kg}$ ,  
 $a = -2.04 \text{ m s}^{-2}$

$$\text{Thus, } a = \frac{0-6}{T-12} = -2 \text{ m s}^{-2}$$

$$T = 15 \text{ s}$$

1M

2

(ii) Height  $\approx$  displacement of lift = area under graph

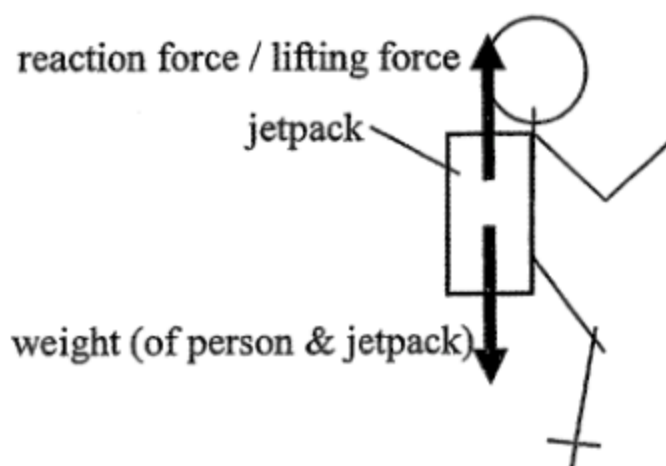
$$= \frac{(12-2)+15}{2} \times 6$$

$$= 75 \text{ m}$$

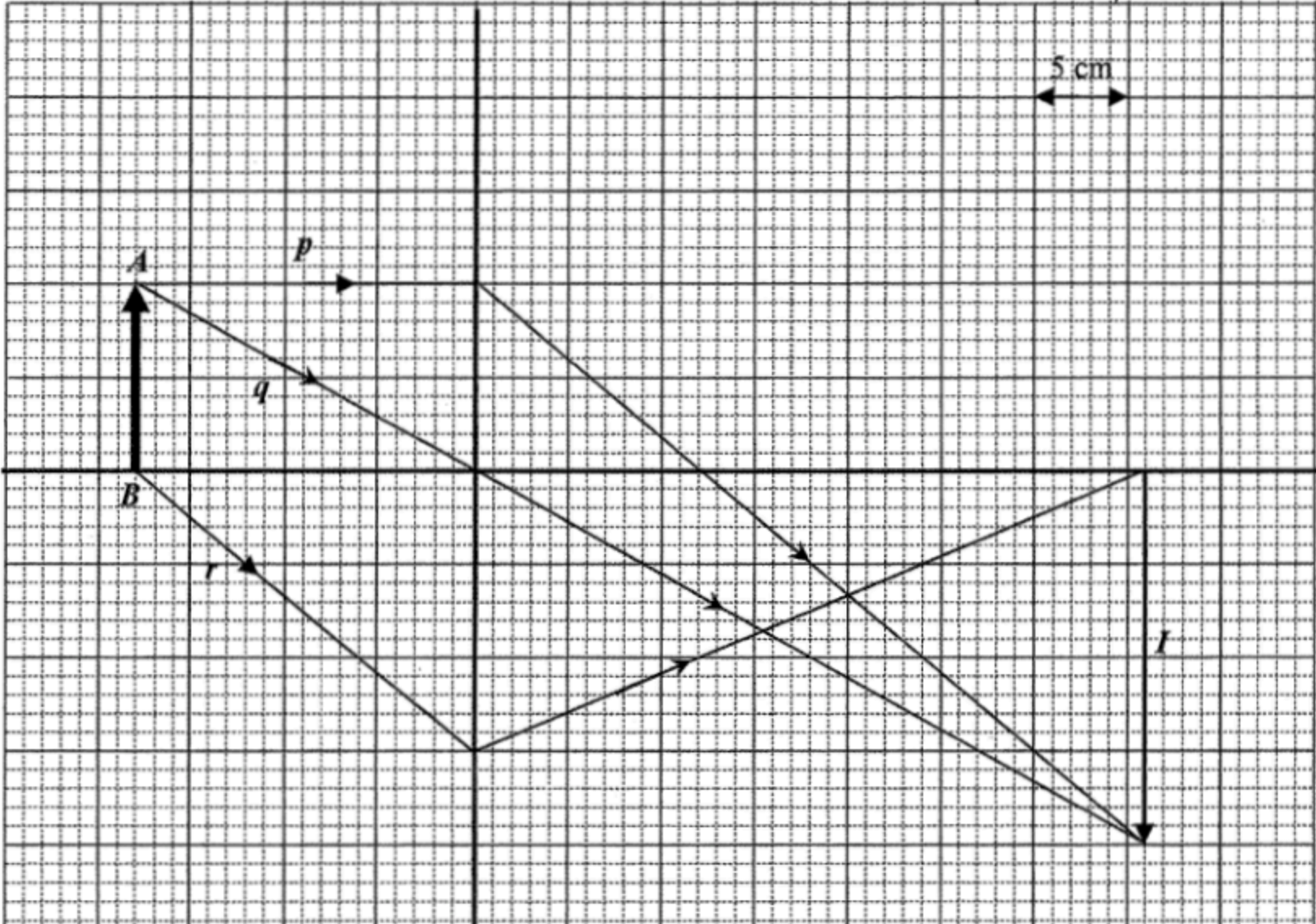
1M

1A

2

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                     |                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>4. (a) According to Newton's second law of motion, a (net) force acts on the water so as to change its momentum. (Or magnitude of the force equals the rate of change of momentum of water).</p> <p>According to Newton's third law of motion, a force acting downwards on the ejecting water (by the jetpack), the water exerts a reaction (equal but upward / opposite) on the jetpack / person as well.</p>                                                                                                                                                                                                                  | <p>1A</p> <p>1A</p> <p>1A</p>                                       |                                                                                                                                                                                                                                    |
| <p>(b)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>3</p> <p>1A</p> <p>1</p>                                         |                                                                                                                                                                                                                                    |
| <p>(c) (i) <math>F = \frac{\Delta p}{\Delta t} = \frac{\Delta m}{\Delta t} \times (\vec{v} - \vec{u})</math></p> <p><math>\frac{\Delta m}{\Delta t} \times (10 - (-10)) \text{ m s}^{-1} = 1000 \text{ N}</math></p> <p><math>\frac{\Delta m}{\Delta t} = 50 \text{ (kg s}^{-1}\text{)}</math></p> <p>(ii) <math>(\frac{\Delta m}{\Delta t})gh + \frac{1}{2}(\frac{\Delta m}{\Delta t}) \times v^2</math></p> <p><math>= (50 \text{ kg s}^{-1})(9.81 \text{ m s}^{-2})(7.5 \text{ m}) + \frac{1}{2}(50 \text{ kg s}^{-1}) \times (10 \text{ m s}^{-1})^2</math></p> <p><math>= 6178.75 \text{ W or } 6.17875 \text{ kW}</math></p> | <p>1M</p> <p>1A</p> <p>2</p> <p>1M</p> <p>1M</p> <p>1A</p> <p>3</p> | <p>Accept using kg as the unit.</p> <p>Accept using <math>m</math> for <math>(\frac{\Delta m}{\Delta t})</math></p> <p>For <math>g = 10 \text{ m s}^{-2}</math>, <math>3750 \text{ W} + 2500 \text{ W} = 6250 \text{ W}</math></p> |
| <p>(d) The same as the same lifting force / (water) jet speed is needed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>1A</p> <p>1A</p> <p>2</p>                                        |                                                                                                                                                                                                                                    |



|                                                                                                                                                                                                                                                                |                       |                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------------------|
| 5. (a) (i) Convex (lens)<br>Only convex lens forms real image (which can be captured by a screen)<br><u>Or</u> Concave lens always forms virtual image (which cannot be captured by a screen).<br><u>Or</u> The image is formed on the other side of the lens. | 1A<br>1A<br>1A<br>1A  |                                |
| (ii)  opaque screen                                                                                                                                                           | 2<br>1A               |                                |
| (b) (i) Image distance $v = 54 - 18 = 36$ cm ( $D = 54$ cm)<br>Magnification = $\frac{v}{u} = \frac{36}{18} = 2$                                                                                                                                               | 1<br>1A<br>1M/1A<br>2 |                                |
| <div style="text-align: center;"><math>L</math></div>                                                                                                                        |                       |                                |
| (ii) Image $I$ of $AB$<br>Rays $p$ and $q$<br>Ray $r$                                                                                                                                                                                                          | 1M<br>1M<br>1M        | 1M for correct position of $I$ |
| (iii) Focal length = $12 \pm 0.5$ cm                                                                                                                                                                                                                           | 3<br>1M/1A            |                                |
| (iv) Move the lens 18 cm farther away from the object.<br><u>Or</u> move the lens 18 cm closer to / towards the screen.<br>Height ratio = 1 : 4.                                                                                                               | 1<br>1M<br>1A<br>2    |                                |

6. (a) (i)

$$\Delta y = \frac{\lambda D}{a}$$

$$\frac{(4.0 - 0) \times 10^{-2}}{10} = \frac{\lambda(1.8)}{0.3 \times 10^{-3}}$$

$$\lambda = 6.666667 \times 10^{-7} \text{ m}$$

$$\approx 6.67 \times 10^{-7} \text{ m or } 667 \text{ nm}$$

1M+1M

1A

3

(ii) To ensure that the light through the 2 slits diffracts enough to interfere / overlap.

1A

1A

Explanation in terms of formula  
 $\lambda = \Delta y \frac{a}{D}$  is NOT accepted as  $a$  is the slit separation given, which is fixed.

2

(b) (i)

$$d \sin \theta = m\lambda$$

$$\frac{10^{-3}}{500} \sin \theta = 6.67 \times 10^{-7} \text{ m}$$

$$\theta = 19.471221^\circ \approx 19.47^\circ$$

Separation between central and first-order bright spot  
 $= 1.8 \tan 19.47^\circ$   
 $= 0.636396 \text{ m} \approx 0.636 \text{ m}$

1M

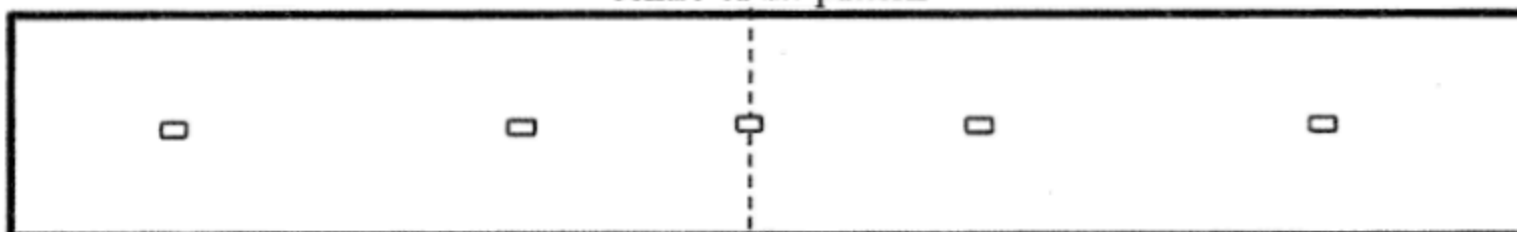
1M

1A

3

(ii)

centre of the pattern



symmetry about the central bright spot (2<sup>nd</sup> order shown)  
 separation between 2<sup>nd</sup> and 1<sup>st</sup> -order bright spots is larger

1A

1A

2

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                       |                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>7. (a) (i) <math>R = 10\text{ k}\Omega</math> (circuit I)</p> $V = \frac{\left(\frac{1}{10\text{ k}\Omega} + \frac{1}{10\text{ k}\Omega}\right)^{-1}}{10\text{ k}\Omega + \left(\frac{1}{10\text{ k}\Omega} + \frac{1}{10\text{ k}\Omega}\right)^{-1}} \times 6\text{ V}$ $= 2\text{ V}$ <p><math>R = 100\text{ }\Omega</math> (circuit II)</p> $V = \frac{\left(\frac{1}{100\text{ }\Omega} + \frac{1}{10\text{ k}\Omega}\right)^{-1}}{100\text{ }\Omega + \left(\frac{1}{100\text{ }\Omega} + \frac{1}{10\text{ k}\Omega}\right)^{-1}} \times 6\text{ V}$ $= 2.985\text{ V}$ | <p>1M</p> <p>1A</p>                   | <p>1M for appropriate method in calculating voltage</p>                                                                                                                                                            |
| <p>(ii) Resistance of circuit / that part of circuit would be lowered / altered significantly when introducing the voltmeter (i.e. loading effect).</p> <p>Or The resistance of the voltmeter is comparable to the resistance of resistor <math>R</math>.</p>                                                                                                                                                                                                                                                                                                                     | <p>1A</p> <p>1A</p>                   | <p>Note: <math>100\text{ }\Omega</math> and <math>10\text{ k}\Omega</math> in parallel <math>\approx 99.0099\text{ }\Omega</math><br/>Accept stating that <math>V</math> slightly <math>&lt; 3\text{ V}</math></p> |
| <p>Resistance of voltmeter should be much higher than the resistance of the part of the circuit under study.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>1A</p>                             |                                                                                                                                                                                                                    |
| <p>(b) (i) <math>V_m</math> does NOT give the true voltage for the resistor.<br/><math>R_m = R_A + R</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>2</p> <p>1A</p> <p>1A</p>          |                                                                                                                                                                                                                    |
| <p>(ii) For circuit III<br/><math>R_m = R + R_A = 10 + 1 = 11\text{ }\Omega</math><br/>percentage error <math>= \frac{1\text{ }\Omega}{10\text{ }\Omega} \times 100\%</math><br/><math>= 10\%</math></p>                                                                                                                                                                                                                                                                                                                                                                          | <p>2</p> <p>1M</p> <p>1A</p> <p>2</p> |                                                                                                                                                                                                                    |



|    |     |      |                                                                                                                               |    |  |
|----|-----|------|-------------------------------------------------------------------------------------------------------------------------------|----|--|
| 8. | (a) | (i)  | - the air loses its insulating properties                                                                                     | 1A |  |
|    |     |      | Or electrons or ions can pass through the air (between clouds and Earth or between clouds and clouds)                         | 1A |  |
|    |     | (ii) |                                                                                                                               | 1  |  |
|    |     |      | $E = \frac{V}{d}$                                                                                                             | 1M |  |
|    |     |      | $V = E d = (3 \times 10^5) \times 2000$                                                                                       | 1A |  |
|    |     |      | $= 6 \times 10^8 \text{ V}$                                                                                                   |    |  |
|    |     | (b)  | (i)                                                                                                                           | 2  |  |
|    |     |      |                                                                                                                               | 1A |  |
|    |     |      | $B = \frac{\mu_0 I}{2\pi r}$<br>$\frac{4\pi \times 10^{-7} \times 30000}{2\pi \times 1500}$<br>$= 4 \times 10^{-6} \text{ T}$ | 1M |  |
|    |     |      |                                                                                                                               | 1A |  |
|    |     |      |                                                                                                                               | 3  |  |
|    |     |      | (ii)                                                                                                                          | 1A |  |
|    |     |      |                                                                                                                               | 1A |  |
|    |     |      |                                                                                                                               | 1A |  |
|    |     |      | (iii)                                                                                                                         | 3  |  |
|    |     |      |                                                                                                                               | 1A |  |
|    |     |      |                                                                                                                               | 1A |  |
|    |     |      |                                                                                                                               | 1A |  |
|    |     |      | Or Lightning current and magnetic field only exist during lightning.                                                          | 1A |  |
|    |     |      |                                                                                                                               | 2  |  |



9. (a) (4) $n_{\alpha} = 238 - 206 \Rightarrow n_{\alpha} = 8$   
 (2) $n_{\alpha} + (-1)n_{\beta} = 92 - 82 \Rightarrow n_{\beta} = 6$

1A

1A

2

(b) (i) 
$$N = N_0 \left( \frac{1}{2} \right)^{t/T_{1/2}}$$

1M

$$\frac{3}{5} = \left( \frac{1}{2} \right)^{t/4.5 \times 10^9 \text{ yr}}$$

Or  $N = N_0 e^{-\lambda t}$  and  $\lambda = \frac{\ln 2}{T_{1/2}}$

1M

$\therefore t = 3.316 \times 10^9 \text{ years} \approx 3.3 \times 10^9 \text{ years}$

1A

2

(ii) Answer in (i) is an underestimate.  
 The original number of U-238 atoms should be greater.

1A

$\therefore$  the ratio  $\frac{\text{present number of U-238 atoms}}{\text{original number of U-238 atoms}} = \frac{N_t}{N_0}$  is

Accept 'more U decayed'.

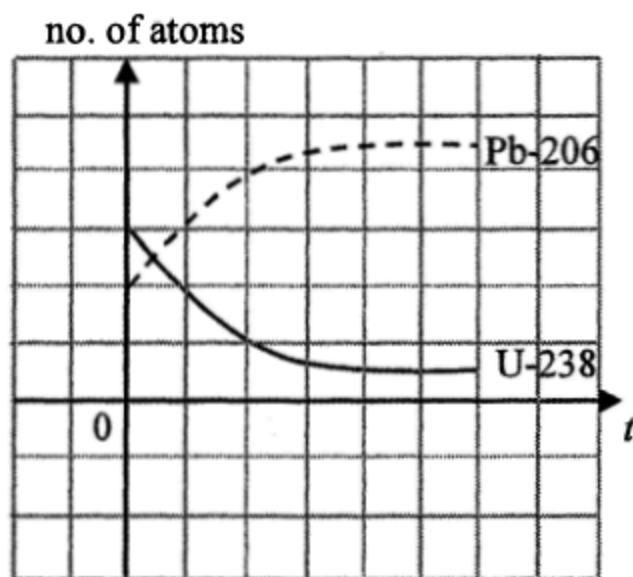
in fact smaller (than  $\frac{3}{5}$ ),

1A

thus longer time should have been elapsed.

2

(iii)



2A

2